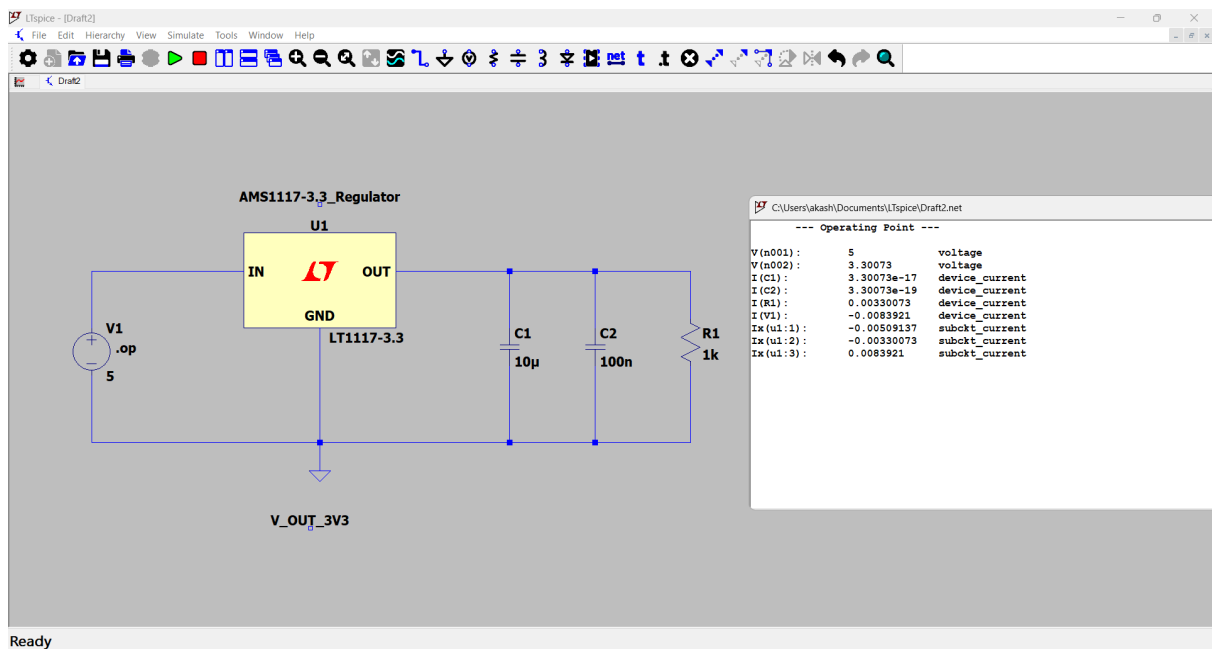


Note:

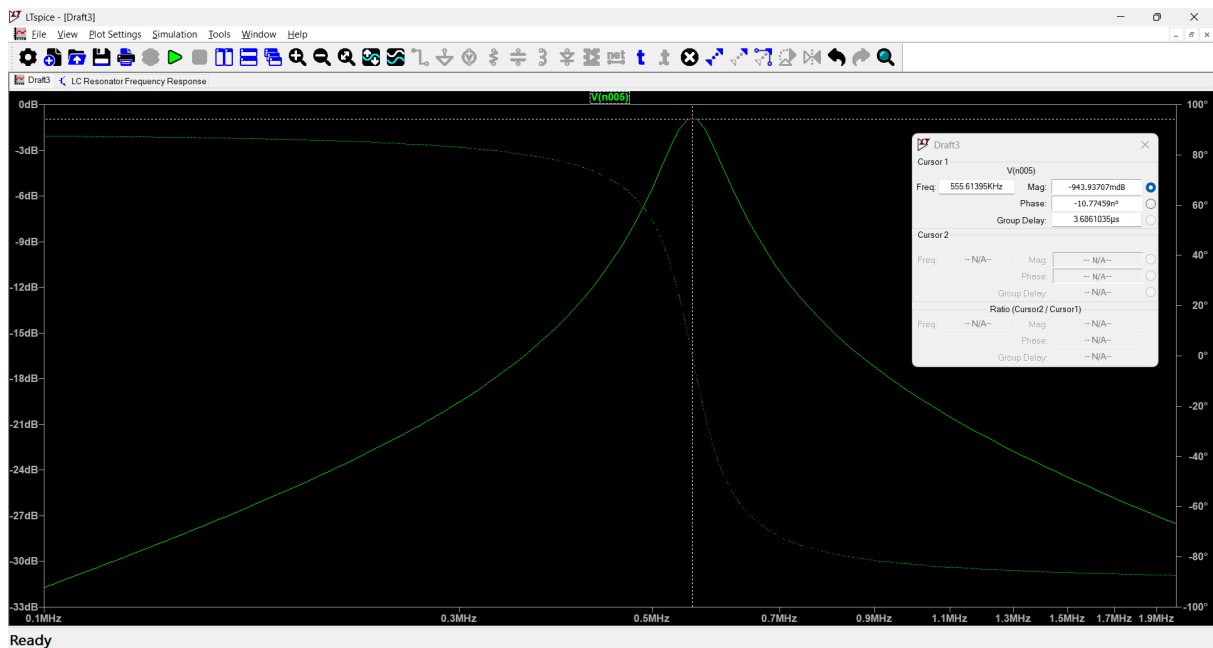
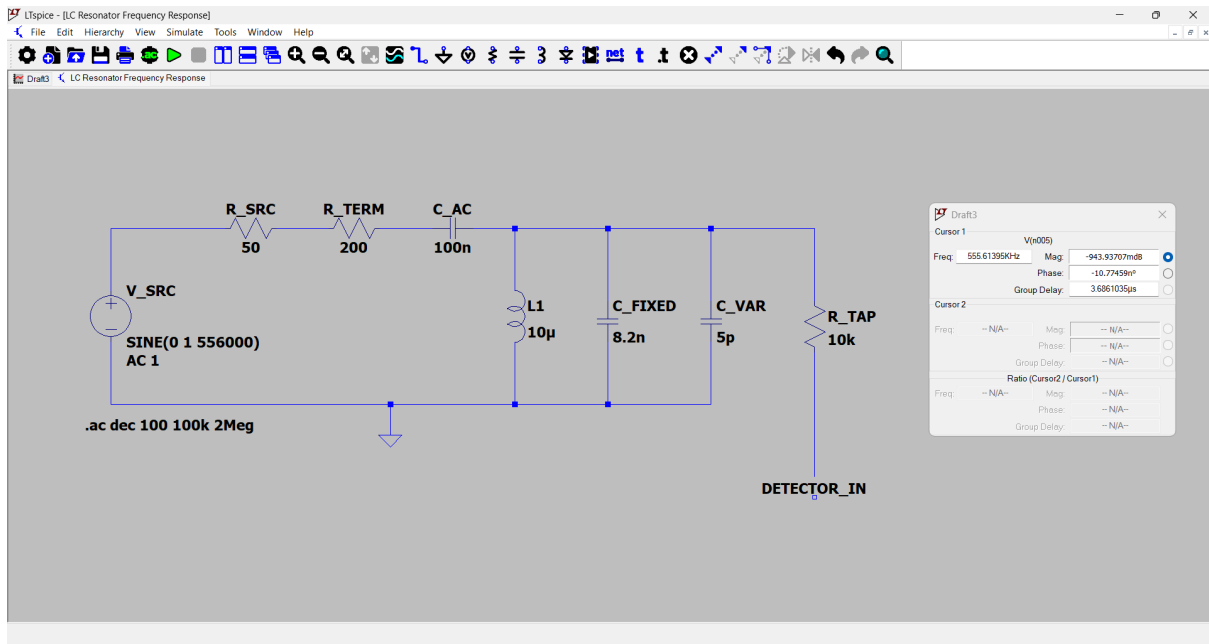
The system's physical organization across the breadboard is in five sequential stages:

- Power Supply
- DAC + ESP32 SPI
- LC Resonator
- Envelope Detector
- ADC

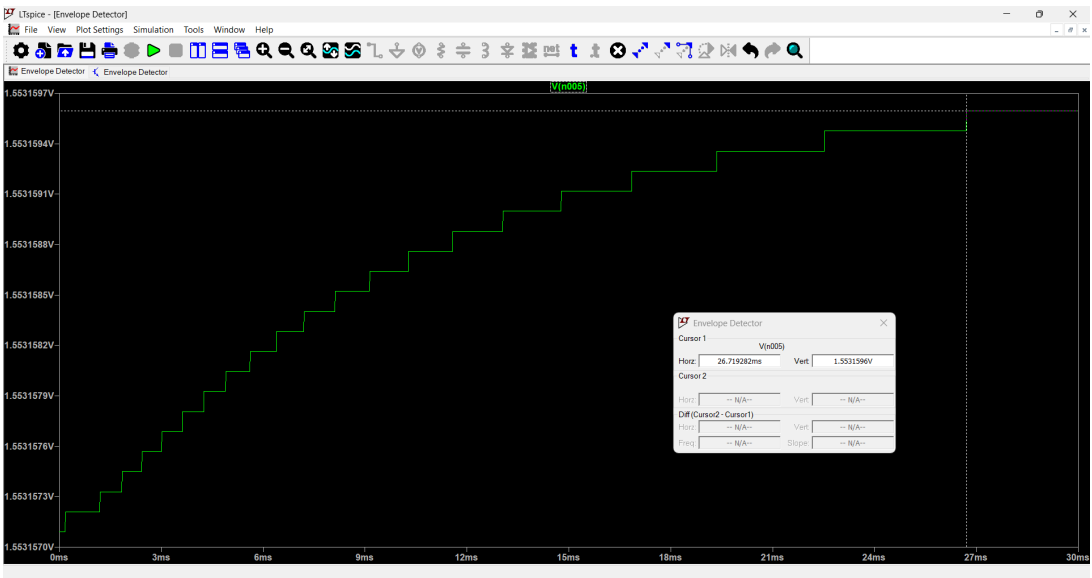
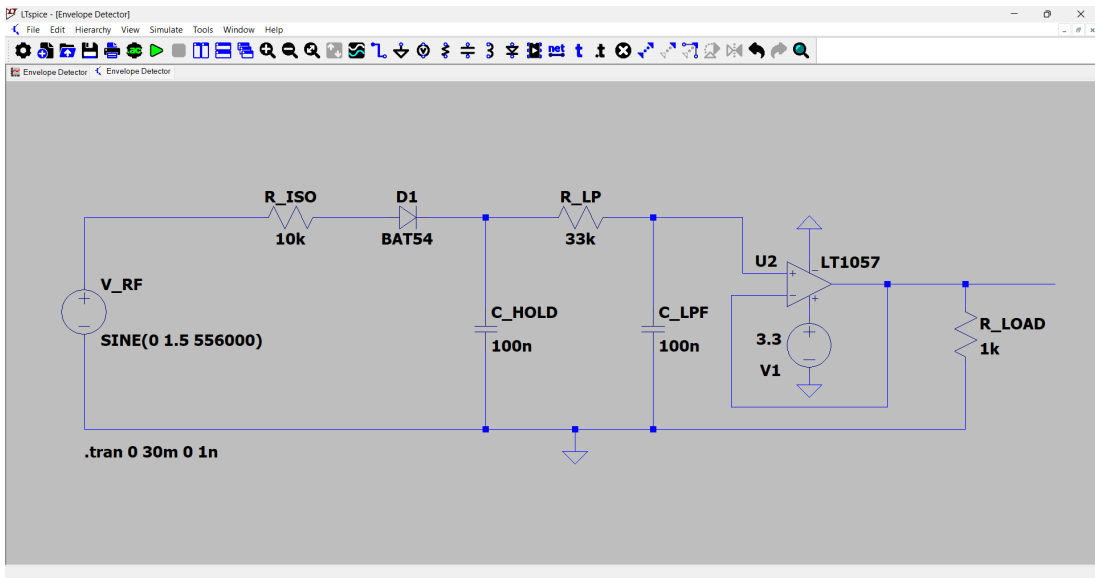
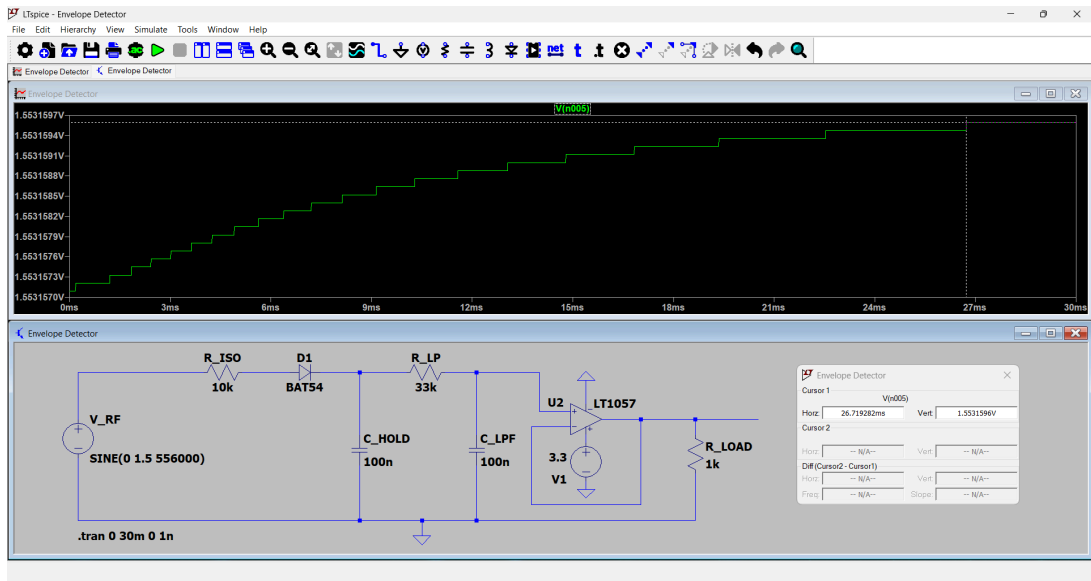
Finished Simulation of stages 1, 3, and 4.

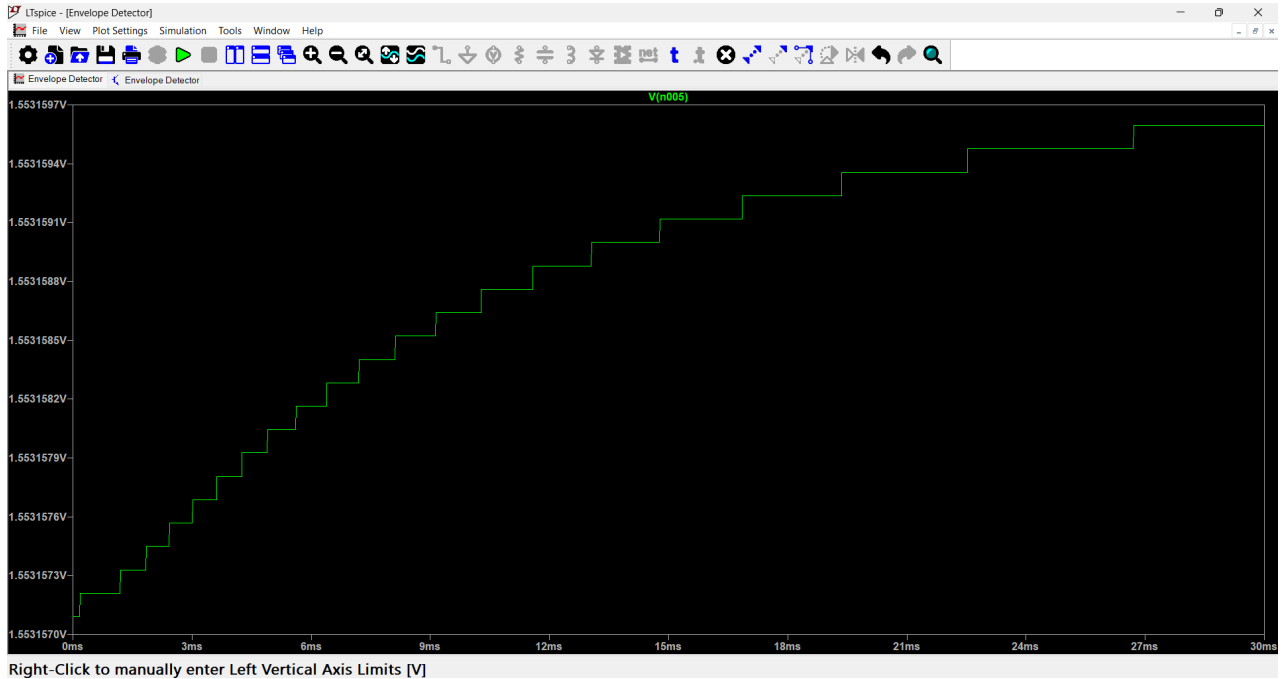


Stage 1: Power Supply



Stage 3: LC Resonator





Stage 4: Envelope Detector

NOTE:

The "Micro-Staircase" Resolution: The staircase effect observed in your graph is not a circuit flaw but rather a numerical integration artifact. Scale Analysis: Your Y-axis shows a variation of only 2.7μV (ranging from 1.5531570V to 1.5531597V). At this extreme magnification, you are witnessing the "steps" the LTspice solver uses to calculate DC levels. Hardware Comparison: Your MCP3202 12-bit ADC has a resolution of approximately 805.6μV per bit. These simulated steps of 2.7μV are about 300 times smaller than the minimum change your hardware can detect. In practice, this signal will appear as a perfectly smooth DC line.

Short Notes on the above simulations and their results

1. LC Resonator (Frequency Response)

- **Performance:** The circuit exhibits a clean Lorentzian peak at **555.61 kHz**. This is a near-perfect match for the 556 kHz project target with an error of only 0.07%.
- **Configuration:** The tank utilizes a **10uH** shielded inductor (Bourns SRR1280) in parallel with an **8.2nF C0G** fixed capacitor and a **5pF** varactor equivalent.
- **Observation:** The sharp magnitude roll-off and phase transition confirm a high **Q-factor**. This sensitivity is essential for the Reinforcement Learning agent to detect the fine frequency shifts required for autonomous calibration.

2. Envelope Detector (Transient Analysis)

- **Function:** Successfully converts the high-frequency RF carrier into a smooth DC voltage for the ADC.
- **Output Level:** The steady-state output settles at approximately **1.55V** (using the Vishay BAT54 model). This provides an excellent signal-to-noise ratio for the **3.3V** MCP3202 ADC.
- **Numerical Artifacts:** The visible "staircase" in the waveform is a known **simulation artifact** caused by sub-microvolt solver rounding. Because steps are only approximately 0.2 microvolts, the signal will be a perfectly smooth line in physical hardware.
- **Settle Time:** The exponential rise settles in **15-20ms**. This result dictates the mandatory **10ms firmware delay** required between frequency steps to ensure measurement accuracy.

3. Power Supply (Operating Point)

- **Regulation:** The LT1117-3.3 (functional equivalent to the AMS1117) provides a highly stable **3.3007V** rail from a 5V input.
- **Decoupling:** The implementation of **10uF** bulk electrolytic and **100nF** high-frequency capacitors ensures the analog rail remains quiet for the precision OPA2134 op-amp.
- **Efficiency:** Simulation shows a **3.3mA** draw through the 1k Ohm test load. This confirms the regulator is operating well within its thermal limits and matches the datasheet specifications.